from these cells is ultracentrifuged in a cesium chloride density gradient. What density distribution of DNA would you expect in this experiment?

(a) One high and one low density band.

(b) One intermediate density band.

(c) One high and one intermediate density band.

**(d) One low and one intermediate density band.**

6. What is the basis for the difference in the synthesis of the leading and lagging strand of DNA molecules?

(a) Origin of replication occurs only at the 5' end of the molecules.

(b) DNA ligase works only in the 3' → 5' direction.

**(c) DNA polymerase can join new nucleotides only to the 3' end of the growing stand.**

(d) Helicases and single-strand binding proteins that work at the 5' end.

7. Which of the following is the correct sequence of event with reference to the central dogma?

(a) Transcription, Translation, Replication

(b) Transcription, Replication, Translation

(c) Duplication, Translation, Transcription

**(d) Replication, Transcription, Translation**

8. Which of the following statements about DNA replication is not correct?

(a) Unwinding of DNA molecule occurs as hydrogen bonds break.

**(b) Replication occurs as each base is paired with another exactly like it.**

(c) Process is known as semi conservative replication because one old strand is conserved in the new molecule.

(d) Complementary base pairs are held together with hydrogen bonds.

9. Which of the following statements is not true about DNA replication in eukaryotes?

(a) Replication begins at a single origin of replication.

(b) Replication is bidirectional from the origins.

(c) Replication occurs at about 1 million base pairs per minute.

**(d) There are numerous different bacterial chromosomes, with replication ocurring in each at the same time.**

10. The first codon to be deciphered was _____ which codes for _____.

(a) AAA, proline (b) GGG, alanine

**(c) UUU, Phenylalanine** (d)TTT, arginine

11. Meselson and Stahl's experiment proved

(a)Transduction

(b) Transformation

(c) DNA is the genetic material

**(d) Semi-conservative nature of DNA replication**

12. Ribosomes are composed of two subunits; the smaller subunit of a ribosome has a binding site for _____ and the larger subunit has two binding sites for two _____. **(mRNA, tRNA)**

13. An operon is a:

(a) Protein that suppresses gene expression

(b) Protein that accelerates gene expression

**(c) Cluster of structural genes with related function**

(d) Gene that switched other genes on or off

14. When lactose is present in the culture medium:

(a) Transcription of *lac y, lac z, lac a* genes occurs.

(b) Repressor is unable to bind to the operator.

(c) Repressor is able to bind to the operator.

**(d) Both (a) and (b) are correct.**



XII Std Zoology Chapter 5 EM.indd 88 23/12/2022 16:07:19